THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 68: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

2. The digit sum of 98765 is: 1

(a) 40 **(b)** 26 **(c)** 35 **(d)** 44

3. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 25 and 47, the number above them is: 1

(a) 72 **(b)** 22 **(c)** 82 **(d)** 1175

4. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 7, 2, 5. The top number is: 1

(a) 16 **(b)** 14 **(c)** 18 **(d)** 28

5. Add 36 and its reverse 63. The sum is: 1

(a) 99 **(b)** 110 **(c)** 121 **(d)** 88

6. Kaprekar step: for 4693, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 6183 **(b)** 6174 **(c)** 6264 **(d)** 6165

7. In the hailstone (Collatz) sequence, the number after 26 is: 1

(a) 52 **(b)** 79 **(c)** 13 **(d)** 14

8. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 79: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

9. The digit sum of 3829 is: 1

(a) 22 **(b)** 19 **(c)** 13 **(d)** 31

10. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 36 and 58, the number above them is: 1

(a) 94 **(b)** 2088 **(c)** 104 **(d)** 22

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Do one Kaprekar step on 4073. 2

MASTERMIND

12. Write the hailstone (Collatz) sequence for 9. 2

13. Apply the reverse-and-add process on 69 until a palindrome appears. 2

14. Do one Kaprekar step on 5381. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the Kaprekar routine on 6352 reaches 6174. 3

16. Find the number of steps taken by the Collatz sequence starting at 17 to reach 1 and the largest number in it. 3

17. A number is formed by writing the digit 4 exactly 9 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

OR

Show that the Kaprekar routine on 8730 reaches 6174. 3

18. Find the number of steps taken by the Collatz sequence starting at 19 to reach 1 and the largest number in it. 3

OR

A number is formed by writing the digit 5 exactly 4 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take the digits of 6021. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 565 and 676 (both palindromes) gives 1241. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 49. 5

20. Take the digits of 8472. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 787 and 898 (both palindromes) gives 1685. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 56. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND